

Minimal Polynomial via Cyclic Vectors

Definition (Annihilator of a Vector)

Let V be a finite-dimensional k -vector space and $f : V \rightarrow V$ linear.

For $v \in V$, the **f -annihilator** of v is the unique monic polynomial $m_v \in k[x]$ of least degree such that

$$m_v(f)v = 0.$$

Remark

Minimality is understood via divisibility:

$$p(f)v = 0; \Rightarrow; m_v \mid p.$$

Definition (Minimal Polynomial)

The **minimal polynomial** of f is the unique monic polynomial $m_f \in k[x]$ of least degree such that

$$m_f(f) = 0.$$

Proposition (Basis Formula)

Let $v_1, \dots, v_n$ be a basis of V . Then

$$m_f = \text{lcm}\{m_{v_1}, \dots, m_{v_n}\}.$$

Proof

Step 1: $\text{lcm} m_{v_i} \mid m_f$

For each i ,

$$m_f(f)v_i = 0.$$

By minimality of m_{v_i} ,

$$m_{v_i} \mid m_f.$$

Hence

$$\text{lcm}\{m_{v_i}\} \mid m_f.$$

Step 2: $m_f \mid \text{lcm} m_{v_i}$

Let

$$P := \text{lcm}\{m_{v_i}\}.$$

It suffices to show $P(f) = 0$. For $v = \sum \alpha_i v_i$,

$$P(f)v = \sum \alpha_i P(f)v_i.$$

Since $P = q_i m_{v_i}$ for each i ,

$$P(f)v_i = q_i(f), m_{v_i}(f)v_i = 0.$$

Thus

$$P(f)v = 0 \quad \forall v \in V,$$

so $P(f) = 0$, hence

$$m_f \mid P.$$

Step 3: Equality

Both are monic and divide each other, hence

$$m_f = \text{lcm}\{m_{v_i}\}.$$

Conclusion

The global invariant m_f is determined by local annihilators on a basis.

Example: Computing m_A

Let

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

Using e_1, e_2, e_3 ,

$$m_A = \text{lcm}\{m_{e_1}, m_{e_2}, m_{e_3}\}.$$

Step 1: e_1

$$Ae_1 = e_1 + e_2, \quad A^2e_1 = -e_1 + 2(e_1 + e_2).$$

Hence

$$A^2e_1 - 2Ae_1 + e_1 = 0.$$

No degree < 2 polynomial works, so

$$m_{e_1} = (x - 1)^2.$$

Step 2: e_2

$$Ae_2 = e_2; \Rightarrow (A - I)e_2 = 0,$$

so

$$m_{e_2} = x - 1.$$

Step 3: e_3

$$Ae_3 = 2e_3; \Rightarrow (A - 2I)e_3 = 0,$$

so

$$m_{e_3} = x - 2.$$

Step 4: LCM

$$m_A = (x - 1)^2(x - 2).$$

Companion Matrix

Definition

For

$$p(x) = x^n - c_{n-1}x^{n-1} + \cdots + c_1x + c_0,$$

define

$$C(p) = \begin{bmatrix} 0 & 0 & \cdots & 0 & -c_0 \\ 1 & 0 & \cdots & 0 & -c_1 \\ 0 & 1 & \cdots & 0 & -c_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -c_{n-1} \end{bmatrix}$$

Proposition

$$m_{C(p)} = p.$$

Proof

Compute

$$C(p)^n e_1 = -c_0 e_1 - c_1 e_2 - \cdots - c_{n-1} e_n.$$

Using $e_{k+1} = C(p)^k e_1$,

$$p(C(p))e_1 = 0.$$

Thus $m_{e_1} = p$, hence

$$m_{C(p)} = p.$$

Remark

e_1 generates V under $C(p)$ (cyclicity).

Nilpotent Operators

Definition

f is **nilpotent** if $f^k = 0$ for some k .

The least such k is the **index**.

Example

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad A^2 = 0.$$

Thus

$$m_A = x^2, \quad \chi_A = x^4.$$

Similarity and Invariants

Proposition

If $A \sim B$, then they share:

- χ_A
 - m_A
 - $\det A$
 - $\operatorname{tr}(A)$
 - $\operatorname{rank}(A)$
-

Example (Failure of Converse)

Let

$$B = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 2 \\ 1 & 1 & 1 & 1 \\ -4 & -4 & -4 & -4 \end{bmatrix}, \quad B^2 = 0.$$

Then

$$\chi_B = x^4, \quad m_B = x^2, \quad \det B = 0, \quad \operatorname{tr}(B) = 0.$$

But

$$\operatorname{rank}(A) = 2 \neq 1 = \operatorname{rank}(B).$$

Conclusion

A and B are **not similar**.

Remark (Jordan Insight)

Equality of χ and m does not fix Jordan structure; rank detects block decomposition.

Summary

- m_f is obtained via

$$m_f = \operatorname{lcm}\{m_v \text{ over a basis}\}.$$

- Companion matrices realize prescribed minimal polynomials.
- Nilpotent maps are classified by their index.
- Invariants constrain similarity, but do not fully determine it.